

A detailed close-up of the front loader assembly of a green Kubota tractor. The image shows the black metal bucket, the hydraulic lift arm, and the green chassis with various hoses and connectors. The tractor is positioned on a light-colored surface.

Keep this manual for later reference.

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Appendix: EC Declaration of Conformity

This manual uses the following warning symbols to indicate factors that must be taken into account to reduce the risk of personal injury or damage to property:

	This safety symbol refers to important safety information in this manual. It warns of an immediate hazard that could cause serious personal injury or damage to property. Read the warning text accompanying the symbol carefully and ensure that other operators are also familiar with the warnings, since personal safety is at stake.
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This notice symbol indicates information about the correct operation and maintenance of the equipment.

Failure to observe the instructions accompanying the symbol can lead to equipment breakdown or other property damage.

In addition to the safety instructions included in this manual, you must observe all occupational safety regulations, local laws and other regulations concerning the use of the equipment. Due to continuous product development, some of the details in this manual may differ from your equipment. We reserve the right to change the contents of the manual without notification. This operator's manual is a translation of the original Finnish version.

1. Foreword

Avant Tecno Oy would like to thank you for your purchase of this AVANT attachment. It has been designed and manufactured on the basis of years of product development and experience. By familiarising yourself with this manual and following the instructions, you ascertain your safety and ensure the reliable operation and long service life of the equipment. Read the instructions carefully before starting to use the equipment or performing maintenance.

The manual includes important instructions for experienced AVANT operators as well. Ensure that all persons using the equipment have received proper guidance and familiarised themselves with the manual and all safety instructions before using the equipment. Keep this manual at hand throughout the service life of the equipment. If the manual is lost, you can request a new one from the retailer or manufacturer. If you sell or transfer the equipment, be sure to hand over this manual to the new owner.

2. Designed purpose of use

The AVANT Grapple bucket is an attachment suitable for use with AVANT multi purpose loaders. The grapple bucket is a universal bucket for various tasks, thanks to its hydraulically operated grapple. The bucket can also be used as a normal bucket with the grapple in the upright position, in transporting various materials, levelling the ground and snow clearing work. The grapple makes it easy to handle loose materials, such as tree branches, chipped wood, compost, manure, straws etc. materials, that would fall from a normal bucket and slip through the silage forks. The grapple of the bucket is not suitable for other purposes than holding material in the bucket.

The attachments have been designed to require as little maintenance as possible. Regular maintenance tasks can be performed by the operator. The assembling of the hydraulic components of the grapple bucket requires some previous experience and basic tools. The assembly and the service of the hydraulic components should be left to qualified professionals, if unsure of how to perform the tasks safely. Appropriate protective equipment and original spare parts must be used for the maintenance. Read the maintenance and servicing instructions in this manual. Please contact your AVANT retailer if you have additional questions about the operation or maintenance of the equipment or if you require spare parts or maintenance services.

3. Safety instructions for the grapple bucket

Please bear in mind that safety is the result of several factors. The loader-attachment combination is highly powerful, and improper or careless use may cause serious personal injury or property damage. Due to this, you should carefully familiarise yourself with the manuals of both the loader and the attachment before starting operation. Do not use the attachment if you have not familiarised yourself with its operation and the related hazards.

The purpose of these instructions is to help you to:

- use the equipment in a safe and efficient manner
- monitor and prevent any hazardous situations
- keep the equipment intact and ensure a long service life



- Avoid impact and crushing hazards that result from falling of the bucket or material falling from the bucket. When attaching a bucket to the loader, ensure that the locking pins of the loader's quick attach plate are in the lower position and that they **have locked the attachment to the loader**.
- Avoid leaking of the hydraulic fluid and possible oil spray. Use the bucket only after it has been installed and inspected correctly. Carefully read instructions concerning installation and assembly.



- The loader may tip over because of overload or uneven terrain. Always transport the load as low as possible to keep the centre of gravity low. The load should be balanced evenly also sideways. Keep the telescopic boom retracted when driving and use it only when reaching to unload the bucket.
- Be especially careful when lifting load to high level or lifting load from high. Operate the controls of the loader calmly and deliberately. When handling heavy loads, avoid sudden changes in speed or direction to maintain balance of the loader.
- To prevent the loader from tipping forward, never leave the driver's seat when the bucket has been elevated. Lower the bucket on the ground.
- Keep the loader articulation in straight position when handling heavy loads. The loader may tilt forward when turning the articulation.
- Use the maximum load indicated by the diagram of the operator's manual of the loader as a guideline. Remember that the actual load carrying capacity varies according to operating conditions, and especially unevenness of operating area has a great effect. Be especially careful when the load sensor indicator is activated.
- Make sure that the operating surface can bear the total load. Also follow the correct tyre pressure settings.
- Lift loads slowly and monitor the position of the bucket during lifting, so that the load will not fall off. Make sure that overhead clearance is sufficient.
- The buckets or the loader are not intended to be left to keep a load elevated for longer periods. It must be observed that the loader boom will lower by itself. Never go under the raised load or the loader boom.
- Pay attention to the contours of the terrain and other hazards, such as branches and trees reaching the driver's area. Pay attention to the surroundings and any other persons and machines moving in the vicinity.
- Shut down the loader engine before any cleaning or maintenance work.
- Make sure that the bucket and its grapple are supported properly before going near the attachment.
- Using buckets for lifting persons or as an access platform is strictly prohibited.
- Do not modify the structure of the attachment in a manner that would affect its safety.
- Also read the safety instructions for the loader in the loader manual.



Note that both the grapple of the attachment and the loader boom move down even if the engine has been stopped (crushing hazard). Never go under a raised boom or the attachment. Lower the grapple and the bucket before leaving the driver's seat.



Familiarise yourself with the loader controls in a safe area. Pay particular attention to the safe stopping of the attachment and the machine.

Remember to wear proper protective equipment and clothing:



- The noise level in the driver's seat may exceed 85 dB(A). Wear hearing protection while working with the loader.



- The use of safety boots is recommended while working with the loader.
- Wear protective gloves when handling hydraulic components.



- Wear safety goggles during hydraulic installations or inspections.

4. Safety instructions for installing hydraulic components

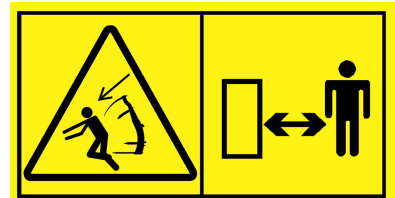
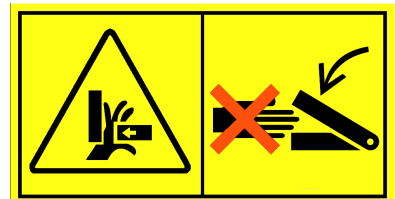
The following safety procedures and general safety practises must be followed when handling hydraulic components. Contact your nearest Avant dealer or service point if you have any additional questions or if you need installation services. It is recommended to leave the installation of hydraulic components to professional service technicians.



High-pressure fluid ejection may penetrate skin and cause serious injuries. Before handling hydraulic components, make sure that the hydraulic system of the attachment and the loader are completely depressurized. Never handle pressurized fittings. See instructions on page 14 to release the residual pressure. Do not hold your hand near a fitting when tightening or opening it. Seek medical attention immediately in case hydraulic fluid is injected through the skin.



Make sure the attachment is securely lowered on the ground. Any part of the attachment must not be supported only by the hydraulic pressure or hydraulic cylinders. Do not loosen any fitting as long as any part of the attachment may move. Use additional supports as necessary. Lower the attachment securely on the ground, shut down the loader engine and lower the grapple on ground or additional supports. Depressurize the hydraulic systems completely.



Ensure that the loader boom or any part of the attachment will not move and cause a crushing hazard:

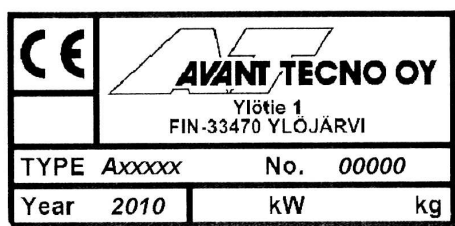
1. Lower the boom to its lowest position
2. Lower the attachment firmly on the ground
3. Lower the grapple or otherwise block the movements of the grapple. Use additional supports or lifting equipment as needed.
4. Shut down the loader engine.
5. Move the auxiliary hydraulics control lever to its extreme positions several times to release residual pressure.

5. Grapple bucket specifications

Listed on the following page are the markings and labels fastened to the attachment, which must be visible on the equipment. Check before use, that the labels are in place and in readable condition. Replace the warning labels, if they have become unclear or detached completely. New labels are available via your retailer or contact information provided on the cover page.



The warning decals contain important safety information. Replace damaged or missing decals with new ones.



Identification plate of the attachment



A46771



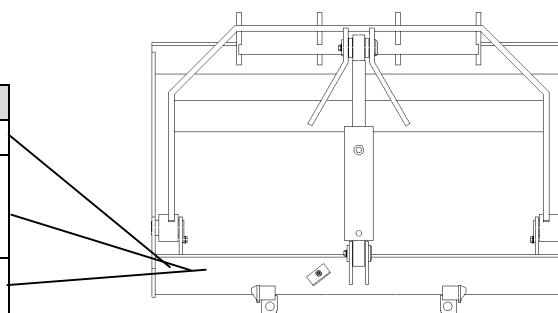
A46772



A46803

Table 1 – Decal warning messages

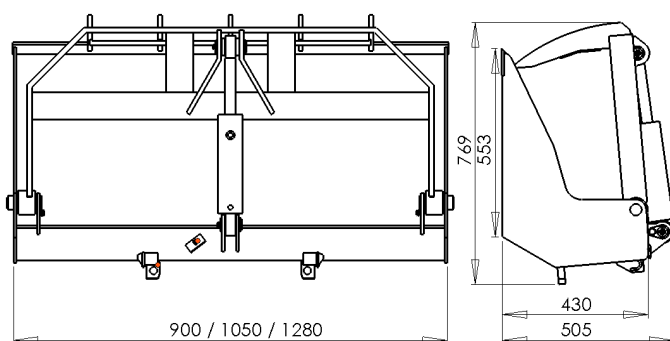
Decal	Explanation
A46771	Read instructions before use.
A46772	Do not go under a raised attachment or the loader boom, keep a safe distance to lifted equipment.
A46803	Crushing hazard, keep clear of moving parts.



Warning decal locations

Table 2 – Grapple bucket technical data

Product number	A32484	A32448	A33639
Working width:	900 mm	1050 mm	1 280 mm
Weight:	130 kg	140 kg	165 kg
Length of the bucket base:	700 mm		
Suitable AVANT loader	300-600	300-700	400-700
Maximum operating pressure:	22,5 MPa (225 bar)		



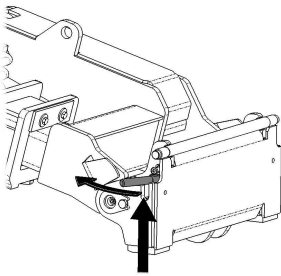
6. Assembly of the grapple bucket

If the hydraulic hoses of the grapple bucket are pre-installed, the bucket is ready to use after attaching it to the loader. In case the hydraulic hoses are delivered as a separate installation kit, follow the instructions shown in section 6.2 to install the hose kit. Read the following instructions concerning the assembly, use and maintenance of the equipment before taking the bucket into use.

6.1. Mechanical attachment of the grapple bucket

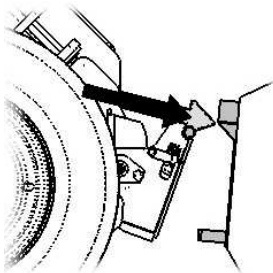
Attaching the attachment to the loader is quick and easy, but it must be done carefully. If the tool is not locked to the loader, it may come off from the loader and cause hazardous situations. In the worst case, the attachment may slide onto the driver along a raised boom. For this reason, the attachment must under no circumstances be raised over one metre when it has not been locked. To prevent hazardous situations, always follow the attachment mounting instructions provided in the following pages. Also remember the safety instructions in Section 3.

The attachment is installed to the loader boom by using the quick attach plate and the counterpart on the attachment. The locking pins are semi-automatic; they will lock when the attachment is tilted towards the loader at low height. The bucket is attached to the loader as follows:



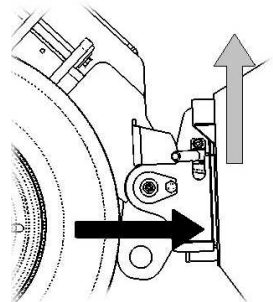
Step 1:

- Lift the quick attach plate locking pin / locking pins up and turn them backwards into the slot so that they are locked in the upper position.
- If your loader is equipped with a hydraulic attachment locking system, see additional instructions about the use of the locking system from the relevant manuals.
- Ensure that the hydraulic hoses are not in the way during installation.



Step 2:

- Turn the quick attach plate hydraulically to an obliquely forward position.
- Drive the loader onto the attachment. If your loader is equipped with a telescopic boom, you can utilise this.

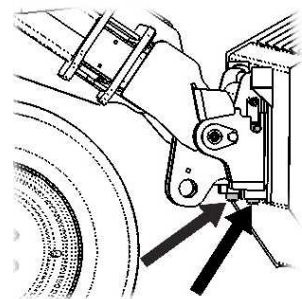


Step 3:

- Lift the boom slightly – pull the boom control lever backward to raise the attachment off the ground.
- Turn the boom control lever left to turn the bottom section of the quick attach plate onto the attachment.
- Lock the locking pins.



Unlocked attachment may fall on the boom or operator and cause injuries. Before moving or lifting the attachment, ensure that the locking pins are in the lower position and come through the fasteners on both sides. Never move or raise an attachment that has not been locked.



Do not use the automatic locking of the locking pins **at a height of over one metre**. If the locking pins do not return to the normal position when the front plate is tilted, do not raise the attachment any more. Lower the attachment on the ground and secure the locking manually.

6.2. Installing the hydraulic hose kit and testing the operation

It is also possible to change the type of fittings by following the procedures in these instructions. Installation kit A45866 includes both multiconnector coupling and conventional quick couplings until the end of 2012.

If the hydraulic hoses are pre-installed to the bucket, check the installation as shown in this chapter. Follow the instructions shown in this manual and general safety procedures of hydraulic assemblies when installing hydraulic components. Also, remember safety instructions from pages 4 and 5. Read through the instructions before beginning installation. Attach the bucket to the loader as shown in previous page before assembling the hydraulic components. Contact your dealer or nearest service point for more information about assembly or installation service.

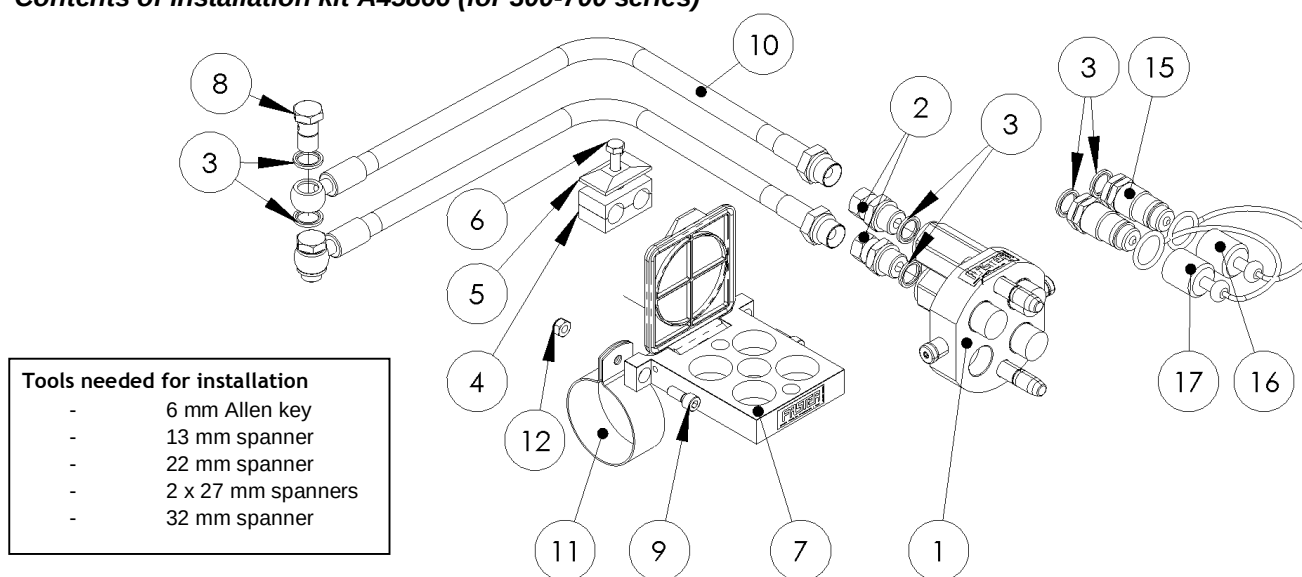


The hydraulic hoses can be assembled correctly only when the attachment has been **attached to the loader**. Keep hydraulic components clean during assembly.



To prevent unintentional movements of the grapple and oil spray, do not connect the quick couplings or the multiconnector to the loader before the hydraulic hoses and all fittings have been tightened. Lower the attachment on the ground, shut down the loader engine, and release any residual pressure from the hydraulic system of the loader. Follow the assembly instructions carefully.

Contents of installation kit A45866 (for 300-700 series)



Tools needed for installation

- 6 mm Allen key
- 13 mm spanner
- 22 mm spanner
- 2 x 27 mm spanners
- 32 mm spanner

A45866 - Hose kit							
Ref.	Part	Qty	Description	Ref.	Part	Qty	Description
1	65251	1	Multiconnector	10	A32277	2	Hydraulic hose L=1900
2	64047	2	Connecting nipple	11	65602	2	Pipe clamp F75
3	73047	8	Usit seal ring R1/2"	12	72320	2	Nut M8 nyloc
4	64887	1	Hose clamp	13	A410342	1	Hose cover spiral
5	73767	1	Cover plate	14	A33633	1	Hose cover spiral
6	73443	1	Hexagonal screw M8*35	15	73409	2	Quick coupling nipple, 300-700 series loaders
7	65594	1	Parking for multiconnector	16	64878	1	Cover, blue, quick coupling
8	73954	2	Banjo bolt R3/8" L29	17	75753	1	Cover, black, quick coupling
9	74481	2	Hexagonal head screw M8*20				

- The assembly kit A44701 may also be used (kit without multiconnector).
- Items 1-3 may be pre-assembled (or the multiconnector 65251 may include fixed fittings)
- Items 15-17 are included in the kit until the end of year 2012.

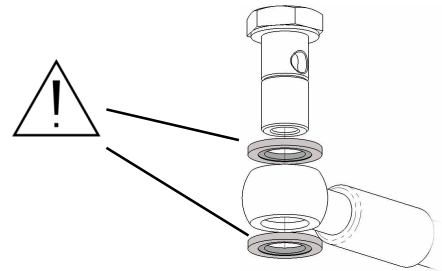
Installing the hydraulic hoses to grapple bucket:

1. Choose a level, well-lit, and clean area for installation. Lower the attachment completely on the ground or on a pallette, shut down the loader engine, and release any residual hydraulic pressure. Also, remember safety instructions from pages 4 and 5.
2. Block the movements of the grapple and carefully remove the protective plastic covers from top of the hydraulic cylinder. Scrape the paint off from the contact surface. Make sure that impurities will not enter the cylinder.



Beware of the possible oil spray from the cylinder. When removing the plastic covers from the cylinder, the grapple will move very easily. As the grapple moves, some oil will be ejected from the cylinder. Use additional supports, e.g. blocks of wood, or lifting equipment, to prevent the grapple from moving inadvertently.

3. The so-called banjo fittings at the end of the hydraulic hoses are connected to the cylinder with hexagonal screw type fittings. Slide a seal ring on both sides of the banjo fitting and tighten to position shown in figures. Note the welding on the cylinder and make sure that the hose is not in contact with the welding.
4. Install the hose clamp to the attachment, but do not tighten the clamp.
5. Install the hydraulic couplings:

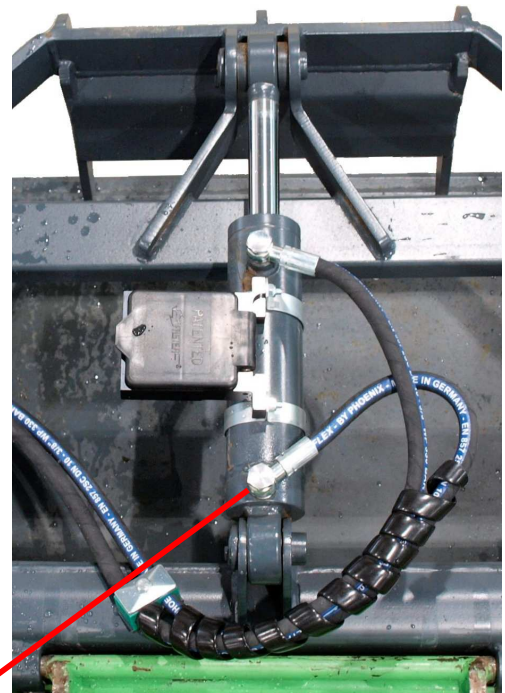


Conventional type quick couplings:

- Slide the protective caps to the straight ends of the hoses. Set a seal ring in between the quick coupling and tighten quick coupling to the hose.
- The coloured cap should be fitted to the hose connected to the piston side of the cylinder, i.e. the fitting closer to the loader.

Installing the multiconnector coupler:

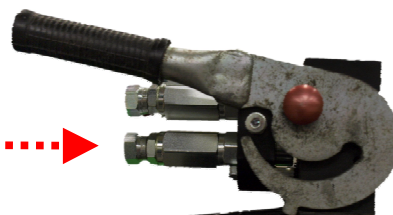
- Install the parking stand of the multiconnector to the hydraulic cylinder.
- Place the multiconnector fitting to the loader connector, but **do not lock the connector to the loader**.
- Fit the connector closer to the loader to the lower fitting of the multiconnector.



Conventional quick couplings



Multiconnector



Do not lock the multiconnector to the loader until the hoses are installed and all fittings have been tightened!



6. A sufficient free length (about 120-130 cm) must be left in between the hose clamp and the hydraulic couplings, so that the attachment can be moved and tilted in all positions of the boom. After installation, test movements of the loader boom as shown on page 11.



7. Set the hoses neatly as shown in adjacent figure, and tighten the hose fittings well.
8. Connect the multiconnector or the conventional couplings to the loader.
9. Coil the protective spiral around the hoses and tighten the hose clamp.
10. Test the tightness of the fittings and the operation of the attachment as shown below.



Tightening hydraulic fittings

Care must be taken when tightening hydraulic fittings, as overtightening of the fittings will break the fittings easily. Tighten fittings tightly using moderate force. Never tighten a fitting if the hydraulic system is pressurized.



Tightening the connecting nipples of the multiconnector:

Lubricate the inner surfaces and threads of the fittings. Turn the tapered JIC-type fittings with fingers or lightly with tools as far as possible. When noticing clear increase in needed force, tighten fitting by turning 1/3 to 1/2 turns using tools. The fitting is not tightened by torque and it will not need a lot of force at any point when tightening. Overtightening will break the fitting. Try to twist the hydraulic hose by hands after tightening, the fitting should stay firm.

Test to find any leaks

- Clean the hydraulic components to find leaks more easily.
- Place a piece of cardboard below the fittings to detect leaks.
- Start the engine. While running on idle, operate the auxiliary hydraulics control lever briefly a couple of times. Shut down the engine, depressurize the hydraulics, and check for any sign of leak. Tighten if necessary.
- Repeat the test by gradually increasing engine rpm and operating time. Use both directions of the auxiliary hydraulics control lever.
- At first the grapple may move inaccurately, as the cylinder and the hoses may have air trapped in them. The system will de-air itself automatically when the grapple is moved to its extreme positions.



Before checking for leakages, or going near any hydraulic fitting, to avoid possible ejection of pressurized hydraulic oil:

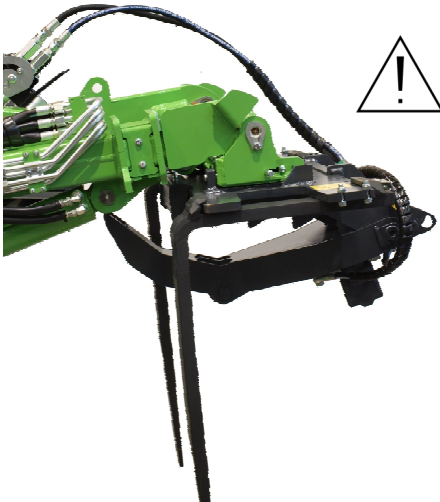
- shut down the loader engine
- depressurize the hydraulic system completely



Test the tightness of the hydraulic fittings carefully. Use idle rpm when starting the first time and keep a safe distance due to the risk of hydraulic fluid ejection. Shut down the loader and release the residual pressure before checking for leaks. Never test the tightness of the hydraulic components by placing hands near the fittings. Never tighten a pressurized hydraulic fitting, since the fitting may break and the released oil may cause serious injuries. Wear eye protection.

Test the movements of the loader boom and the attachment

After installing and testing the hydraulic components, check that the hydraulic hoses can not get stretched or pinched in any position of the loader boom or the attachment.



Carefully test the movements of the loader boom. Release the hydraulic pressure before the test. Adjust the free length of the hoses as necessary.

6.3. Connecting and disconnecting the hydraulic couplings



Never connect or disconnect quick couplings or other hydraulic components if the control lever of the auxiliary hydraulics control lever is locked on or while the system is pressurized. Connecting or disconnecting the hydraulic couplings while the system is pressurized may lead to unintended movements of the attachment or ejection of high-pressure fluid, which can cause serious injuries.

Keep the fittings as clean as possible; use the protective caps for the quick couplings on both the attachment and the loader. Dirt, ice, etc. may make using the fittings significantly more difficult. Never leave the hoses hanging on the ground; place them on top of the equipment.

Connecting standard quick couplings:

Before connecting or disconnecting the standard quick couplings, shut down the loader engine and move the auxiliary hydraulics lever on the loader to its extreme positions a few times. This way, there will be no residual pressure in the system and it is possible to connect the fittings.

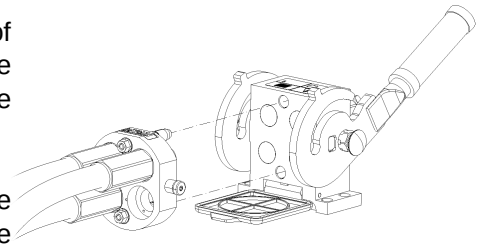
To connect and disconnect the standard couplings, move the collar at the end of the female fitting. The hoses should be connected so that the coupling equipped with a coloured cap is connected to the corresponding coupling of the loader. Note that the protective caps on the loader and the attachment can be fastened to each other during operation to reduce the accumulation of dirt. When disconnecting the standard quick couplings a small amount of oil may drip from the couplings. Wear protective gloves and have some cloth at hand to keep the equipment clean.



Connecting the multiconnector system:

Align the pins of the attachment connector with corresponding holes of the loader connector. The multiconnector will not connect if the attachment connector is upside down. Connect and lock the multiconnector by turning the lever towards the loader.

The lever should move easily all the way to its locking position. If the lever does not slide smoothly, check the alignment and position of the connector and clean the connectors. Also shut down the loader and release the residual hydraulic pressure.



Disconnecting the hydraulics:

Before disconnecting the fittings, lower the attachment on solid and level surface. Turn the control lever of the auxiliary hydraulics to its neutral position.



When removing the attachment, always disconnect the hydraulic couplings before unlocking the quick attach plate, to prevent hose damage and any oil spills. Reinstall the protective caps on the fittings to prevent impurities from entering the hydraulic system.

Releasing residual hydraulic pressure:

In case residual pressure is left in the hydraulic system of the attachment, it is often possible to disconnect the hydraulic couplings, but it may be difficult to connect them the next time. If the fittings will not connect, the residual pressure must be released by turning the auxiliary hydraulics control lever of the loader, when the engine is turned off. To make sure that there will not be residual pressure in the hydraulic system of the attachment, shut down the loader engine and move the auxiliary hydraulics control lever of the loader before disconnecting the couplings.

6.4. Storage of the attachment

When finished working with the grapple bucket lower the attachment on an even surface where it cannot move by accident. Lower the grapple completely first.

Do not store the attachment directly against the ground; place it on blocks of wood or on a pallet, for example. Do not leave the attachment to a position where water will stay in the bucket. Before longer storage periods touch up the paint where needed, grease the lubrication points and oil the visible parts of hydraulic cylinder shafts to prevent rust damage.

7. Operation

The bucket is ready for use after it has been attached, installed and tested according to chapters 5 and 6. Practise the use of the attachment and the controls of the loader in a safe area. If you are unfamiliar with the loader to be used, it is recommended to practice its use without any attachments.

- Before use check that the loader and the bucket are in good condition and that all obstacles, including any hidden ones, have been removed from the working area. Remember correct working methods and avoid leaving the driver's seat unnecessarily.
- Ensure that the loader and the attachment are used in a safe manner and as intended. Operate the loader controls only when sitting at the driver's seat. Do not allow children to operate the machine.
- Check regularly that the attachment is intact and to ensure there are no hydraulic leaks. Refer to Section 7 for maintenance instructions.



Ensure that others are at a safe distance when operating the equipment. Do not allow others near the boom danger area or to stay directly in front or behind of the loader.

Instructions for using the bucket

Avoid overturning of the machine. Do not extend the telescopic boom when the load approaches the lifting capacity of the loader or when the load has been lifted high. When the load indicator of the loader signals, lower the load calmly and retract the telescopic boom. Avoid sudden changes in speed or direction and drive especially carefully on inclined or uneven terrain. Use additional counterweights as needed.



Lifting material from a pile requires the most power from the loader. With correct working methods loading of material becomes more efficient, since ramming into a pile and then attempting to tilt the bucket is not possible with wide buckets and heavy materials. Loading is most efficient when only the tip of the bucket is pushed into a pile, and then with simultaneous driving, tilting and lifting the bucket is filled with material. This way the bucket will get completely filled the easiest way. Use high engine rpm.



Avoid hazards of falling materials. In case the loader is not equipped with the boom self levelling, the front plate must be tilted simultaneously to prevent the load from falling from on the loader or the operator. Fitting the loader with the boom self-levelling option makes the repeated lifting work easier.

When transporting objects with the grapple bucket, ensure that the load is secured on the attachment. Use load-securing straps if the use of the grapple is not possible to keep the load in the bucket.



Do not use the buckets as a work platform or to transport persons even temporarily.



When lifting material from a high level, the loader may tip forward, when starting to reverse with the loader. Do not reverse and drag the load simultaneously, but make sure, that the loader can handle the load being moved.



Avoid hazards related to sudden stopping of the loader. Hitting an obstacle will stop the loader suddenly, for example in snow clearing work. Check the operating area before operation and use the seat belt.

A boom floating system is available for some loader models. Note that the floating will not prevent the loader from stopping suddenly in case an object has been hit.

7.1. Operating on inclined terrain

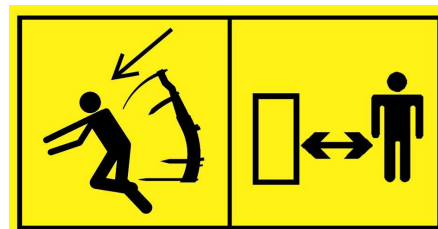
Using the loader on inclined terrain or slopes requires extra caution. Maximum lifting capacity can not be achieved on inclined terrain. On horizontally tilted terrain the load should not be lifted to high levels. The frame articulation should be kept straight when lifting heavy loads; turning the load during lifting operation will affect the stability of the loader and may lead to turning over of the machine. Drive slowly and avoid sudden changes in speed or direction. Operate the boom with careful and calm movements.

8. Maintenance and servicing

Continuous maintenance of the grapple bucket includes regular cleaning and lubrication. The condition of the hydraulic hoses and components should be monitored regularly. Due to crushing hazard caused by the grapple, all maintenance work must be done when the grapple has been lowered down completely and the hydraulic fittings are uncoupled.



The grapple may lower by itself as a result of a fault or use or service mistake causing serious injuries. All maintenance procedures must be performed when the grapple has been lowered down completely.



Check the condition of the hydraulic hoses and components when the engine has been turned off and the pressure has been relieved. Repair all leaks immediately after detecting them; a small leak can quickly grow into a big one. Do not use the equipment, if you have discovered a fault in the hydraulic system. Leaking hydraulic fluid may penetrate the skin and cause serious injuries. Hydraulic fluid is also harmful to the environment.

Check hoses visually for cracks or abrasions. Monitor the wearing of the hoses and stop the use, if the surface layer of the hose has worn off. If there are signs of leaking, to check the component, hold up a piece of cardboard in the area where a leak is suspected. Finding any of these faults means that the hydraulic hose or component in question must be replaced. Spare parts are available from your nearest AVANT retailer or service point.



Never tighten a hydraulic fitting under pressure, since the fitting may break and the released oil may cause serious injuries. Do not use the equipment, if you have discovered a fault in the hydraulic system.



8.1. Cleaning the equipment

Clean the grapple bucket regularly to prevent accumulation of dirt that is more difficult to remove. You may use a pressure washer and mild detergent. Do not use strong solvents or spray directly at the hydraulic components or at the labels on the attachment either.

8.2. Lubricating

The grapple bucket has a total of four lubrication points, to which a small amount of lubricant is added regularly. The lubrication points are located at the both ends of the hydraulic cylinders and two more are at the grapple pivot. All lubrication nipples are standard R1/8" nipples. Replace any damaged nipples.



Lubricant must be added approximately every 1 to 3 times a year, depending on usage. Clean the end of the nipple before greasing. Only add the grease amount of a couple of grease press pushes at a time.

9. Warranty terms

Avant Tecno Oy grants a warranty of one year (12 months) for the attachment from the date of purchase.

The warranty covers repair costs as follows:

- Work costs are covered, if the repair is not performed at the factory.
- The factory replaces any defective components or consumables.

The factory may reimburse the price of components purchased by the customer in special cases that have been agreed in advance.

The warranty does not cover:

- Normal maintenance work, such as the replacement of blades or the belts, or parts and consumables required for it.
- Damages caused by unusual operating conditions or ways of use, negligence, overload, structural changes made without the consent of Avant Tecno Oy, use of non-original parts or lack of maintenance.
- Consequences of a defect, such as interruption of work or other possible additional damages.
- Travel and/or freight costs caused by the repair.

EY-vaatimustenmukaisuusvakuutus
EG-försäkran om överensstämmelse
EC Declaration of Conformity
EG-Konformitätserklärung



Valmistaja / Tillverkare / Manufacturer / Hersteller:

AVANT TECNO OY

Osoite / Adress / Address / Adresse:

Ylötie 1

33470 YLÖJÄRVI, FINLAND

Vakuutamme täten, että alla mainitut tuotteet täyttävät konedirektiivin turvallisuus- ja terveysvaatimukset (direktiivi 2006/42/EY muutoksineen). Seuraavia yhdenmukaistettuja standardeja on sovellettu:
 SFS-EN ISO 12100-1, SFS-EN ISO 12100-2, SFS-EN 982, SFS-EN 474-1

Vi försäkrar härmed att nedan beskrivna produkter överensstämmer med hälso- och säkerhetskrav i EG-maskindirektiv (EG-direktiv 2006/42/EG som ändrat). Följande harmoniserade standarder har tillämpats:
 SFS-EN ISO 12100-1, SFS-EN ISO 12100-2, SFS-EN 982, SFS-EN 474-1

We hereby declare that the products listed below are in conformity with the provisions of the Machinery Directive (directive 2006/42/EC as amended). The following harmonized standards have been applied:
 SFS-EN ISO 12100-1, SFS-EN ISO 12100-2, SFS-EN 982, SFS-EN 474-1

Wir erklären hiermit, dass die nachstehend aufgeführte Maschine konform ist mit den Bestimmungen der EG-Maschinenrichtlinie (EG-Richtlinie 2006/42/EG mit Änderungen). Die folgende harmonisierte Normen wurden angewandt:

SFS-EN ISO 12100-1, SFS-EN ISO 12100-2, SFS-EN 982, SFS-EN 474-1

Kuvaus: Avant-kuormainten kanssa käytettävä hydraulitoiminen kauha
 Beskrivning: Hydraulisk skopa; arbetsredskap för Avant lastare
 Description: Hydraulic bucket; attachment for Avant loaders
 Beschreibung: Hydraulische Schaufel; Anbaugerät für Avant Radlader

Mallit / Modeller / Models / Modellen:

Nimi / Namn/ Name / Name		Leveys Bredd Width Breite
Kombikauha	A32484	90 cm
Grepskopa	A32448	105 cm
Grapple bucket	A33639	130 cm
Greifschaukel		

Allekirjoitus / Namnteckning / Signature / Unterschrift: _____ 10.6.2010 Ylöjärvi, Finland

Risto Käkelä,
 Toimitusjohtaja / Verkställande direktör /
 Managing Director / Geschäftsführer





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